Rule-Based Medical Symptom Checker

Overview

This project is a simple **AI-based expert system** developed using Python. It helps users identify possible medical conditions based on the symptoms they enter. The system works by comparing user symptoms with a predefined knowledge base of diseases and calculating the best match.

This project was created as part of a first-year engineering course on **Fundamentals of Artificial Intelligence and Machine Learning**, focusing on concepts from Modules 1–3 such as knowledge representation and rule-based reasoning.

Features

- Accepts symptoms from the user through a simple interface
- Uses rule-based logic to match symptoms with diseases
- Displays a ranked list of possible conditions
- Shows the most probable disease based on match percentage
- Implemented using only basic Python concepts

Technologies Used

- Python 3
- Jupyter Notebook

No external libraries or frameworks were used, making the project easy to understand and run on any system with Python installed.

How the System Works

The project is based on three main components:

1. Knowledge Base

A dictionary is used to store diseases and their associated symptoms.

2. Inference Engine

The program compares user-entered symptoms with stored symptoms and calculates a match percentage for each disease.

3. Output Module

The system displays a diagnosis report and highlights the most probable condition.

How to Run the Project

Step 1: Install Python

Make sure Python 3 is installed on your system.

Step 2: Open Jupyter Notebook

Launch Jupyter Notebook and create a new Python notebook.

Step 3: Copy the Code

Paste the full project code into a single cell.

Step 4: Run the Program

Run the cell and enter symptoms when prompted, for example:

fever, cough, headache

The system will display possible diseases and their match percentages.

Sample Output

```
========== Diagnosis Report ==========


Flu: 75.00% match

Malaria: 50.00% match

Common Cold: 33.33% match


Most probable condition: Flu
```

Project Structure

```
Medical-Symptom-Checker/
│
├── symptom_checker.ipynb
├── README.md
└── project_report.pdf
```

Limitations

- The system uses a small set of diseases and symptoms
- It is designed for educational purposes only
- It does not replace professional medical advice

Future Improvements

- Add more diseases and symptoms
- Store patient history
- Create a graphical user interface
- Improve accuracy using machine learning techniques